# K.R.Sagara Dam Water Conservation Assistant

## Layer A Technical Stabilization — Final Engineering Completion Report

**Date:** 2026-11-05
**Author / Role:** Antigravity AI Assistant (Google DeepMind)
**Target Contract:** antigravity_layer_a_implementation_plan.md
**Scope:** Layer A Only (Layer B Excluded)
**Test Suite Status:** 46 Passed / 0 Failed
**Execution Environment:** Python 3.11.9 (.venv)

**Executive Summary:** All non-blocked tasks specified in the Layer A Implementation Plan have been successfully executed, tested, and verified. The KRS Dam dataset engineering, unit conversions, storage variable semantics, water-balance equations, dynamic forecasting freshness, and dashboard have been stabilized to ensure technical consistency, strict reproducible MLOps provenance, and zero false leakage claims. Layer B (recommendations, allocation optimization, gate control) remains strictly unbuilt.

## 1. Summary of Files Changed & Engineering Rationale

| File Path | Action | Purpose & Engineering Rationale |
|---|---|---|
| **config/reservoir_reference.yaml** | Created | Centralized reservoir identity, capacity references (Design Gross 49.452 TMC, Live 45.051 TMC, 2009 Survey 42.99 TMC), and unverified live denominator policy. |
| **config/data_quality.yaml** | Created | Centralized physical plausibility bounds (elevation 2500 ft, storage 49.452 TMC, non-negative flows) and NaN preservation policies. |
| **config/water_balance.yaml** | Created | Centralized unit conversion factor (0.0000864 TMC/cusec-day), unverified evaporation status, interval matching rules, and anti-leakage disclaimers. |
| **config/freshness.yaml** | Created | Configurable observation delay threshold (1 day) and dynamic freshness labeling criteria. |
| **config/rainfall_stations.yaml** | Created | Documents rainfall pipeline blocker and 10 regional proxy stations awaiting verified external data ingestion. |
| **src/utils/unit_conversions.py** | Created | Pure, centralized flow rate (cusecs) to volume (TMC) conversions over arbitrary hours with strict negative/NaN validation. |
| **src/utils/validate_conversions.py** | Created | Validates daily converted flow against cumulative meter deltas across 36 years; exports validation report. |
| **src/data/validate.py** | Enhanced | Generates CSV/JSON audit reports; detects duplicates, negative values, missing dates; flags anomalies without deleting raw records. |
| **src/models/storage_analysis.py** | Created | Pure functions distinguishing gross vs live storage, enforcing continuous daily intervals (blocking multi-day deltas), and setting unverified percentages to N/A. |
| **src/models/water_balance.py** | Created | Canonical mass balance module returning explicit result contracts; enforces interval alignment; blocks numeric discrepancy when evaporation is unverified. |
| **src/models/water_budget.py** | Refactored | Maintained as backward-compatible wrapper delegating to unit_conversions.py and water_balance.py. |
| **src/data/preprocess.py** | Refactored | Eliminated zero-filling of missing evaporation; enforces 1-day interval for storage change diff; uses pure unit conversions. |
| **src/models/predict.py** | Enhanced | Adds dynamic observation freshness calculation, prevents relabeling stale historical records as 'tomorrow', verifies model metadata integrity. |
| **src/app.py** | Stabilized | Modernized dashboard with dynamic data freshness badges, unverified denominator warnings, water-balance diagnostics, and clear leakage disclaimers. |

## 2. Baseline Problems Confirmed From Actual Code

• **Evaporation Imputation:** src/data/preprocess.py previously filled 546 missing evaporation readings with 0.0, masking physical data gaps.

• **Unverified Evaporation Units:** Correlation between daily evaporation volume and cumulative evaporation delta is only 0.5108 (vs 1.000000 for inflow/outflow), proving source convention is unverified.

• **Broken Storage Change Across Calendar Gaps:** gross_capacity_tmc.diff() assumed daily rows; across missing date 2020-02-11, it evaluated a 2-day change and compared it to 1-day flow.

• **Storage Semantics Conflation:** Gross storage was conflated with live storage, and percentage full was computed against gross design capacity without bathymetric verification.

• **Scattered Magic Numbers:** The constant 0.0000864 was duplicated across multiple modules without input validation.

• **Stale Forecast Presentation:** The 15-day-old forecast (origin 2026-10-21) was displayed as 'tomorrow's forecast' without dynamic freshness verification.

## 3. Engineering Assumptions & Domain Rationale

• **Daily Mean Flow Factor:** 1 cusec sustained for 24 hours equals exactly 0.0000864 TMC (validated by 1.000000 correlation against cumulative meter deltas).

• **Dead Storage Sill Offset:** Difference between gross capacity and live storage above cill is approximately 4.401 TMC (49.452 - 45.051 = 4.401 TMC).

• **Evaporation Incompleteness Policy:** Unverified evaporation convention blocks complete mass-balance closure; discrepancy is held at None for unverified intervals.

• **Live Storage Denominator Policy:** Storage percentage is reported as N/A / unverified until certified bathymetric surveys are provided.

• **Freshness Policy:** Forecasts are marked HISTORICAL_STALE if the latest observation is older than 1 day relative to current system execution.

## 4. Unresolved Source & Domain Questions

• **Evaporation Measurement Methodology:** Does Evaporation (Cusecs) represent pan evaporation with an undocumented factor, or reservoir depth loss without dynamic surface area weighting?

• **Gross Capacity Readings Above Design:** On 1994-10-06, 1999-02-26, and 2011-02-14 to 2011-02-16, recorded values peak at 69.322 TMC (exceeding 49.452 TMC design capacity). Historical log verification from Karnataka WRD is required.

• **Current Hydrographic Live Storage Bathymetry:** An updated sedimentation survey since 2009 is required to establish a certified live capacity denominator.

## 5. Flow Unit Conversion & Water Balance Verification

Unit conversions were validated across 13,442 daily historical records against cumulative flow meters:

| Flow Series | Valid Pairs | Correlation | MAE (TMC) | Status | Engineering Finding |
|---|---|---|---|---|---|
| **River Inflow** | 13,396 | 1.000000 | 0.000319 | **PASS** | Extreme consistency; validates 0.0000864 TMC/cusec-day factor. |
| **River Outflow** | 13,403 | 1.000000 | 0.000339 | **PASS** | Extreme consistency with cumulative outflow meter. |
| **Canal Withdrawal** | 13,403 | 0.999993 | 0.000364 | **PASS** | High consistency with cumulative withdrawal meter. |
| **Evaporation** | 12,857 | 0.510785 | 0.000197 | **UNVERIFIED** | Low correlation; convention and area-elevation curve unverified. |

### Historical Water-Balance Audit Status (13,441 Evaluated Intervals)

- **UNVERIFIED_EVAPORATION:** 12,888 intervals (95.88%) — Aligned flow terms complete, but numeric discrepancy blocked due to unverified evaporation convention.
- **INCOMPLETE:** 553 intervals (4.12%) — Incomplete due to missing evaporation readings or missing calendar day (2020-02-11).
- **Anti-Leakage Disclaimer:** Departures from the simplified daily mass balance reflect unmeasured catchment runoff, localized rainfall, travel timing delays, or gauge calibration; they do NOT prove dam leakage.

## 6. Chronological Inflow Forecaster Benchmarks

Models were trained strictly on pre-2023 data (1990–2022, 12,035 observations) with zero future target leakage, selected on validation data, and benchmarked on held-out test data:

| Model / Baseline | Partition | MAE (cfs) | RMSE (cfs) | $R^2$ | WAPE (%) |
|---|---|---|---|---|---|
| **Persistence Baseline (t-1)** | Validation (2023-2024) | 2,168.32 | 8,111.96 | 0.5634 | 48.51% |
| **7-Day Moving Average** | Validation (2023-2024) | 2,682.49 | 8,639.46 | 0.5049 | 60.01% |
| **Ridge Regression** | Validation (2023-2024) | 2,246.33 | 7,610.15 | 0.6152 | 50.25% |
| **HistGradientBoosting** | Validation (2023-2024) | 2,185.74 | 7,539.02 | 0.6224 | 48.90% |
| **Tuned Random Forest (Winner)** | Validation (2023-2024) | 2,060.03 | 7,192.17 | 0.6558 | 46.08% |
| **Persistence Baseline (t-1)** | Held-Out Test (2025-2026) | 2,278.40 | 6,685.56 | 0.4796 | 43.20% |
| **Tuned Random Forest** | Held-Out Test (2025-2026) | 2,347.07 | 5,929.95 | 0.5906 | 44.50% |

**Test Benchmark Finding:** Tuned Random Forest achieves an **11.30% reduction in RMSE** over persistence on held-out test data, with $R^2$ improving from 0.48 to 0.59.

## 7. Layer A Gate Completion Matrix & Layer B Boundaries

| Gate / Scope Element | Status | Implementation Details & Justification |
|---|---|---|
| **Gate 0: Baseline Inspection** | DONE | Internal inventory captured; baseline report written to reports/layer_a_baseline.md. |
| **Gate 1: Centralized Data Validation** | DONE | Schema, duplicates, calendar gaps, and plausible ranges audited in reports/data_quality_report.csv. |
| **Gate 2: Storage Semantics** | DONE | storage_analysis.py created; live vs gross storage separated; unverified percentage set to N/A. |
| **Gate 3: Flow Unit Conversions** | DONE | unit_conversions.py implemented; 1.000000 correlation validated in unit_conversion_validation.csv. |
| **Gate 4: Water Balance Correction** | DONE | Canonical water_balance.py enforces interval alignment; unverified evaporation blocks discrepancy. |
| **Gate 5: Rainfall Pipeline** | BLOCKED | Raw rainfall source archive absent from repository; documented transparently in config and reports. |
| **Gate 6: Forecasting Integrity** | DONE | Dynamic freshness checks in predict.py flag archive age (15 days stale); target derived from origin. |
| **Gate 7: Anomaly Separation** | DONE | Separated data-quality, water-balance, and residual flags; eliminated all leakage claims. |
| **Gate 8: MLOps Reproducibility** | PARTIAL | Local DVC pipeline and MLflow tracking fully functional; cloud remote blocked (no external credentials). |
| **Gate 9: Dashboard Stabilization** | DONE | src/app.py stabilized with dynamic freshness badges, storage semantics, and verified outputs. |
| **Layer B: Recommendation Engine** | FORBIDDEN / NOT IMPLEMENTED | No recommendation APIs, tables, or recommendation ML models created per Section 0 contract. |
| **Layer B: Water Allocation Optimization** | FORBIDDEN / NOT IMPLEMENTED | No gate-dispatch, demand optimization, or release scheduling logic added. |
| **Layer B: Leakage Diagnosis** | FORBIDDEN / NOT IMPLEMENTED | Discrepancies explicitly disclaimed as unmodeled runoff or timing delays; never leakage. |

## 8. Reproduction Commands & Automated Test Verification

All 46 focused Layer A tests pass with zero failures:

```
# 1. Run complete Layer A automated test suite
.\.venv\Scripts\pytest tests/test_validation.py tests/test_storage_analysis.py tests/test_unit_conversions.py
 tests/test_water_balance.py tests/test_forecasting.py tests/test_inference_freshness.py tests/test_units_and
_budget.py tests/test_app_smoke.py -v

# 2. Run data validation and export CSV/JSON audits
.\.venv\Scripts\python.exe -m src.data.validate

# 3. Run flow unit conversion validation
.\.venv\Scripts\python.exe -m src.utils.validate_conversions

# 4. Launch stabilized Streamlit dashboard
.\.venv\Scripts\python.exe -m streamlit run src/app.py
```